

AETHERA · SPECIALTY BILLING SERIES

Cardiology Billing Solutions

Maximizing Revenue Across E/M, Cardiovascular Sessions, Procedural Billing, Devices, and Cardiac Rehab

Prepared by: Aethera Healthcare Solutions · Lakeland, FL

Scope: CPT 93000–93299 · 93452–93653 · 93797–93798 · Device interrogations

Updated: 2026 Edition · aetherahealthcare.com · +1 (863) 694-0325





ABOUT AETHERA

Maximizing Revenue. Minimizing Burden.

Aethera Healthcare Solutions is your full-service medical billing partner headquartered in Lakeland, FL. We handle coding, claims, payments, denials, and collections — so you can focus on patients.

500+

Practices Served

across 25+ medical specialties

\$12M+

Revenue Recovered

denied & underbilled claims recouped

95%

Clean Claim Rate

vs 84% industry avg

< 30d

Average AR

days in accounts receivable



Personal, Not Corporate

Dedicated account team with deep specialty/payer knowledge.



Performance Targets, In Writing

96%+ net collection, <5% denial, 95%+ coding accuracy, 24h posting, 48h submission, 65%+ appeal success.



Zero Risk to Start

No setup fees, no long-term contracts, 90-day cancel, free 5-day audit.



CARDIOLOGY · MARKET CONTEXT

Why Cardiology Billing Matters Now

Cardiology generates the highest code density per encounter in outpatient medicine — every patient visit averages 3-6 separately billed services, each with its own documentation, modifier, and pre-auth requirements.



\$216B

US cardiovascular disease treatment spend (2024)

Largest single disease category.



121M

Americans with some form of cardiovascular disease

Includes hypertension, the most common chronic condition.



200+

Distinct cardiovascular CPT codes

Spanning ECG, holter, stress, cath, EP, devices, and rehab.



~ 22%

Cardiology claim denial rate (industry avg)

Highest of any outpatient specialty, driven by modifier errors and missing pre-auth.



Net effect: cardiology practices that master modifier -26/-TC splits, document device interrogations completely, and pre-auth every procedure can recover **6-12% of total revenue** currently lost to coding errors and denials.



CARDIOLOGY · OVERVIEW

The Four Pillars of Cardiology Billing

Each pillar represents a distinct service family with unique coding, modifier, and pre-auth requirements.

01 · OFFICE E/M + SESSIONS



Office E/M & Cardiovascular Sessions

Office visits (99202–99215) plus diagnostic cardiovascular sessions: ECG (93000), holter monitoring (93224–93227), event monitors (93241–93248), stress testing (93015).

◆ CPT 99202–99215 · 93000–93010
· 93224–93272

02 · PROCEDURES



Cardiac Procedures

Catheterization (93452–93461), coronary intervention (92920–92944), electrophysiology studies and ablation (93600–93653). High-dollar procedural billing with strict pre-auth.

◆ CPT 92920–92944 · 93452–
93461 · 93653

03 · DEVICES



Cardiac Device Management

Pacemaker and ICD interrogations, remote monitoring (93294–93299), loop recorders (93298–93303). Recurring revenue per patient per year.

◆ CPT 93294–93299 · 93298–93303

04 · REHAB



Cardiac Rehabilitation

Supervised exercise sessions for post-MI, post-CABG, post-stent, and heart failure patients. 36-session Medicare benefit; codes 93797 (initial) and 93798 (subsequent).

◆ CPT 93797 · 93798 · 36
sessions max



PILLAR 01 · OFFICE E/M + SESSIONS

Diagnostic Sessions & Office Visits

Cardiology office encounters typically combine an E/M visit with one or more cardiovascular diagnostic sessions. The 2021 E/M coding reform simplified documentation (MDM-based, not history-based) but added complexity around which sessions can be billed separately on the same day. Modifier 25 is required when billing E/M + procedure on the same encounter.

E/M coding (2021+ reform)

Code by Medical Decision Making complexity or time; 99202–99215. Modifier 25 required if same-day procedure.

ECG (93000–93010)

93000 = complete ECG with interpretation. 93005 = tracing only. 93010 = interpretation only. Cannot bill both 93005 and 93010.

Stress testing

93015 = complete stress test (monitoring + interpretation + supervision). 93016–93018 components billable separately when not all performed by same provider.



Cardiologists average **4.2 billable services per encounter** — proper modifier use is the single biggest revenue determinant.

CPT Quick Reference

2026 national average reimbursement · professional component

Code	Service	Avg \$
99213	Established, low MDM, 30-39 min	\$92
99214	Established, moderate MDM, 40-54 min	\$131
99215	Established, high MDM, 55-69 min	\$187
93000	Complete ECG with interpretation	\$18
93015	Cardiovascular stress test, complete	\$116



PILLAR 02 · PROCEDURES

Cath, Intervention, and Electrophysiology

Cardiac procedures represent the highest-dollar outpatient cardiology services. Reimbursement ranges from \$425 for a diagnostic left heart cath to \$2,800+ for an EP ablation. Pre-auth is mandatory for all cath and EP procedures; missing pre-auth is the #1 denial reason.

01

Diagnostic Catheterization

93452 left heart cath (\$425); 93458 coronary with imaging (\$575); 93460 left + right heart cath (\$625); 93463 right heart cath only (\$415).

02

Coronary Intervention (PCI)

92920 single vessel balloon angioplasty (\$750); 92928 single vessel drug-eluting stent (\$850); 92929 addl vessel DES (\$450); 92941 acute MI stent (\$950).

03

Electrophysiology

93600 comprehensive EP study (\$1,200); 93653 ablation for arrhythmia (\$2,800); 93620 inducibility study (\$800).

04

Structural Heart

93312 TEE guidance (\$425); 33407 TAVR (separate surgical code); 93662 intracardiac echo (\$475).



Pre-Auth Requirements

All cath and EP procedures require pre-auth from commercial payers. Medicare does NOT require pre-auth for cath. Typical PA turnaround: 5-10 business days.



Modifier -26 vs -TC

Professional component (-26) billed by interpreting cardiologist. Technical component (-TC) billed by facility. Cannot bill both unless same provider owns both.



Bundling Rules

CATH + PCI: 93458 bundled into 92920; do not bill separately. EP + ablation: 93600 bundled into 93653; do not bill separately.



PILLAR 03 · DEVICE MANAGEMENT

Pacemaker, ICD, and Loop Recorder Monitoring

Cardiac implantable electronic devices (CIEDs) — pacemakers, ICDs, and implantable loop recorders — require ongoing monitoring for the life of the device (typically 5-10 years). Monitoring generates recurring revenue per patient per quarter, but most practices under-bill because they don't document device checks properly or miss the quarterly cycle.

In-person interrogation

93294 pacemaker in-person check (\$58); 93296 ICD in-person check (\$98). Billed once per 90 days.

Remote monitoring

93295 pacemaker/ICD remote interrogation (\$58); 93296 remote monitoring programming (\$120). Billed once per 91 days.

Implantable loop recorder

93298 in-person analysis (\$120); 93299 remote monitoring (\$120); 93303 monitoring + data review (\$225). Quarterly cycle.



A practice with 1,000 implanted patients generates ~ **\$232K/year** of device monitoring revenue at near-zero marginal cost — but most practices capture only 60-70% of eligible monitoring cycles.

\$58

Per pacemaker in-person check

CPT 93294

\$178

Per ICD quarterly monitoring cycle

93295 + 93296

\$225

Per loop recorder quarterly cycle

CPT 93303

~ \$232/yr

Avg annual device monitoring revenue per patient

across CIED types

+ PILLAR 04 · CARDIAC REHAB

Supervised Exercise for Post-Event Recovery

Cardiac rehab is a CMS-covered supervised exercise + education program for patients who have had: MI, CABG, valve surgery, heart transplant, PCI/stent, stable angina, or heart failure (EF ≤35%). Medicare covers up to 36 sessions per lifetime event; commercial payers typically follow.

◆ 01 · INITIAL VISIT

Initial Visit (93797)

Comprehensive initial assessment: medical history, exercise tolerance test, individualized treatment plan. Billed once per rehab episode.

▣ 02 · SUBSEQUENT SESSIONS

Subsequent Sessions (93798)

Supervised exercise session with ECG telemetry monitoring; typically 60 minutes. Billed per session, up to 36 sessions.

🛡️ 03 · ELIGIBLE DIAGNOSES

Eligible Diagnoses

MI (410.x0), CABG (414.02), valve surgery (V42.2), PCI/stent (414.01), heart failure (428.x), angina (413.x), post-transplant (V42.1).

📊 04 · SESSION STRUCTURE

Session Structure

2-3 sessions/week × 12-18 weeks. Telemetry monitored; BP before/after; symptoms documented. Minimum 30 min aerobic exercise.

INITIAL VISIT (93797)

Per-session reimbursement

\$55

SUBSEQUENT SESSION (93798)

Per-session reimbursement

\$48

MAX SESSIONS

Per Medicare benefit period

36

ANNUAL REVENUE PROJECT

50-patient panel × 36 sess

\$86,400 / yr · per 50 active

⚠️ COMPLIANCE NOTE

Documentation required per session: telemetry strip, BP before/after, exercise duration, symptoms. Missing telemetry strip is the #1 denial reason.

CARDIOLOGY · BILLING CODES

Top CPT Codes & 2026 National Average Reimbursement

Most-billed codes across the four cardiology pillars. Professional component shown (modifier -26 for facility billing).

SOURCE · CMS PFS 2026
NATIONAL AVG · NON-
FACILITY

Code	Service	Modifier	Avg \$	Notes
99213	Est. patient, low MDM, 30-39 min	—	\$92	Most common outpatient E/M code
99214	Est. patient, moderate MDM, 40-54 min	—	\$131	2nd most common; cardio avg visit level
99215	Est. patient, high MDM, 55-69 min	—	\$187	Complex cardio management
93000	ECG, complete with interpretation	—	\$18	Often same-day as E/M (modifier 25)
93015	Cardiovascular stress test, complete	—	\$116	Monitoring + interp + supervision
93224	Holter monitor, 24hr, scan + interp	-26	\$290	Up to 48hr add 93225
93241	Event monitor, 30-day, analysis	-26	\$190	Patient-activated; longer than holter
93452	Left heart cath, diagnostic	-26	\$425	Pre-auth required commercial
93458	Coronary cath with imaging	-26	\$575	Bundled w/ PCI (92920)
92920	Coronary balloon angioplasty, single	—	\$750	Add -LC/-LD for vessel
92928	Coronary DES, single vessel	—	\$850	Add -LC/-LD/-RC for vessel
93653	EP ablation for arrhythmia	—	\$2,800	High-dollar; pre-auth mandatory

COMPONENT & SAME-DAY MODIFIERS

-26 = Professional component (interpretation only). -TC = Technical component

PCI VESSEL MODIFIERS

-LC = Left circumflex. -LD = Left anterior descending. -RC = Right coronary. Used with PCI



🔗 CARDIOLOGY · DENIAL PREVENTION

Top 8 Cardiology Denial Reasons & How Aethera Prevents Each

Cardiology denial rates average 18-22% — highest of any outpatient specialty. Each denied claim costs \$50-200 to rework given procedural complexity.

● RED · 01

24 %

Missing pre-auth

ROOT CAUSE: Cath, EP, PCI performed without prior authorization on file

AETHERA: Auto-submits PA at scheduling; verifies approval within 5 days of procedure

● RED · 02

19 %

Modifier -26/-TC errors

ROOT CAUSE: Professional and technical components billed incorrectly; double-billed

AETHERA: Modifier rules engine enforces correct -26/-TC split per provider/facility setup

● AMBER · 03

15 %

Missing vessel modifier

ROOT CAUSE: PCI code (92920/92928) submitted without -LC/-LD/-RC vessel identifier

AETHERA: PCI documentation checklist requires vessel modifier before claim release

● AMBER · 04

12 %

Bundling violation

ROOT CAUSE: Diagnostic cath (93458) billed separately from PCI (92920) on same encounter

AETHERA: Bundling rules engine blocks unbundled claims at submission

● AMBER · 05

9 %

Same-day E/M without modifier 25

ROOT CAUSE: E/M + procedure on same day without modifier 25; procedure bundled

AETHERA: Auto-appends modifier 25 when E/M + procedure codes co-occur

● GREEN · 06

7 %

Device interrogation documentation missing

ROOT CAUSE: 93294/93295 billed without interrogation report or remote transmission log

AETHERA: Device documentation checklist requires transmission ID before billing

● GREEN · 07

6 %

Cardiac rehab telemetry strip missing

ROOT CAUSE: 93798 billed without telemetry strip showing monitored exercise

AETHERA: Rehab documentation checklist requires telemetry strip upload

● GREEN · 08

5 %

Stress test components unbundled

ROOT CAUSE: 93016/93017/93018 billed separately when same provider performed all

AETHERA: Bundling rules enforce 93015 when single provider performs all components



CUMULATIVE IMPA

Aethera reduces cardiology denial rate from industry avg 20 % to < 8 % — recovering ~ \$180K/year per \$1M practice revenue.



■ CASE STUDY · INTEGRATED CARDIOLOGY RCM

4-Cardiologist Practice — Annual Revenue Impact

A \$4.2M-revenue practice models the combined upside of modifier discipline, device monitoring capture, and pre-auth workflow automation. 12,000 active patients · 1,500 implanted device patients · in-office cath lab.

⚙️ LEVER

Modifier Discipline

Annual procedural claim volume	3,800
Pre-Aethera modifier error rate	18%
Post-Aethera modifier error rate	4%
Recovered claims	532 / yr
Avg procedural claim value	\$625

ANNUAL REVENUE

\$332,500

♥️ LEVER

Device Monitoring Capture

Implanted device patients	1,500
Pre-Aethera cycle capture	65%
Post-Aethera capture	92%
Recovered cycles / year	1,215
Avg cycle value (ICD)	\$178

ANNUAL REVENUE

\$216,270

✓ LEVER

Pre-Auth Automation

Annual procedural volume req. PA	1,200
Pre-Aethera PA denial rate	22%
Post-Aethera PA denial rate	6%
Recovered procedures	192 / yr
Avg procedural revenue	\$1,850

ANNUAL REVENUE

\$355,200



Combined annual revenue recovery: \$903,970. Implementation cost (year 1): ~ \$260K (Aethera RCM fees + workflow retraining + cath lab pre-auth integration). Year-1 net: \$643,970. ROI: 248%.



⚖️ CARDIOLOGY · COMPLIANCE HOTSPOTS

Where OIG, Medicare RAC, and Commercial Payers Audit First

Top compliance risks across the four cardiology pillars. Quarterly self-audits recommended.

PILLAR E/M + Sessions 99202-99203 93000-93009	PILLAR Procedures 92920-92929 93452-93459	PILLAR Devices 93294-93299 93298-93300	PILLAR Cardiac Rehab 93797-93799 SESSION
RED Modifier 25 misuse — appending to E/M codes when procedure on same day does not represent separately identifiable service; OIG Work Plan 2025-2026.	RED Cath + PCI unbundling — billing diagnostic cath (93458) separately from PCI (92920); bundled and not separately billable.	HIGH Quarterly cycle fraud — billing 93294/93295 more than 4 times per year per patient; CMS quarterly limit enforcement.	RED Missing telemetry documentation — billing 93798 without telemetry strip showing monitored exercise; automatic denial on post-pay audit.
AMBER Stress test component unbundling — billing 93016+93017+93018 separately when same provider performed all; should be bundled to 93015.	RED Missing pre-auth documentation — performing cath/EP without payer pre-auth on file; commercial payers deny at 100%.	MEDIUM Remote monitoring without transmission — billing 93295 without documented remote transmission log from device manufacturer.	AMBER Eligible diagnosis not documented — patient enrolled in rehab without ICD-10 diagnosis matching Medicare coverage list.
GREEN Same-day ECG + E/M without separate documentation — ECG included in E/M if not separately documented.	AMBER Wrong vessel modifier — using -RC when left circumflex (-LC) treated; medical record must support vessel.	LOW In-person vs remote miscode — billing 93294 (in-person) when service was remote; should be 93295.	LOW Exceeding 36-session limit — billing 93798 beyond Medicare lifetime benefit without separate medical necessity documentation.



OIG Work Plan 2025-2026 explicitly lists modifier 25 misuse in cardiology and cardiac device monitoring cycle fraud as active audit targets. Medicare RACs have recovered **> \$180M** from cardiology practices since 2020.



Three Actions to Capture Cardiology Revenue

For cardiology practice administrators and revenue cycle directors leading the next 90 days.

01



Deploy a modifier rules engine

Audit the last 200 procedural claims for modifier errors (missing -26/-TC, missing vessel -LC/-LD/-RC, missing 25 on E/M+procedure days). Implement pre-submission rules that block claims with missing modifiers.

TARGET: **< 5% modifier error rate** (industry avg 18%)

02



Build a device monitoring capture workflow

Inventory all 1,500+ implanted device patients. Build quarterly monitoring calendar with auto-alerts at 91-day cycle mark. Track transmission logs to ensure 100% capture. Each missed cycle = \$178+ in pure lost revenue.

TARGET: **92%+ capture rate**

03



Automate pre-auth for every cath/EP procedure

Integrate EHR scheduling with payer pre-auth portals. Auto-submit PA at time of scheduling. Track 5/10/15-day turnaround SLAs. Pre-auth denial rate drops from 22% to <6% — recovering ~\$355K per 1,200 annual procedures.

TARGET: **< 6% PA denial rate**



Questions & Discussion

Optimizing Cardiology Revenue Across E/M, Procedures, Devices & Rehab



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